## Exercice 2 :

- by linearity of $\mathbb{E}$, $\mathbb{E}[Z_t] = \mathbb{E}[X_t + Y_t]$

$$= \mathbb{E}[X_t] + \mathbb{E}[Y_t]$$

$$\boxed{\mathbb{E}[Z_t] = \mu_x + \mu_y}$$

where $\mu_x + \mu_y$ are the constant by stationarity of $\{X_t\}$ and $\{Y_t\}$

- for $h \in \mathbb{N}$,

$$\text{cov}(Z_t, Z_{t-h}) = \text{cov}(X_t + Y_t, X_{t-h} + Y_{t-h})$$

by bi-linearity of cov we have

$$\text{cov}(Z_t, Z_{t-h}) = \text{cov}(X_t, X_{t-h}) + \underbrace{\text{cov}(X_t, Y_{t-h})}_{=0,\ \{X_t\} \perp \{Y_t\}} + \underbrace{\text{cov}(Y_t, X_{t-h})}_{=0\ \{X_t\} \perp \{Y_t\}} + \text{Cov}(Y_t, Y_{t-h})$$

then $\boxed{\text{cov}(Z_t, Z_{t-h}) = \gamma_x(h) + \gamma_y(h)}$

by stationnarity of $\{X_t\}$ and $\{Y_t\}$ where $\gamma_x$ and $\gamma_y$ are the autocovariance of $\{X_t\}$ (resp $\{Y_t\}$) depend only on the lag $h$

- conclusion : $\{Z_t\}$ is stationary because
  - $\mathbb{E}[Z_t] = \mu_Z$ (constant of time)
  - $\text{cov}(Z_t, Z_{t-h}) = \gamma_Z(h)$, $\forall h \in \mathbb{Z}$